World-R1: Reinforcing 3D Constraints for Text-to-Video Generation

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Research Questions

- How can video foundation models be enhanced with 3D geometric consistency without architectural modifications or expensive inference-time constraints?
- Can reinforcement learning with 3D foundation model feedback effectively align video generation with world-modeling requirements?
- How can camera trajectory control be achieved implicitly through noise conditioning rather than explicit architectural components?

Core Idea

Video foundation models excel at visual synthesis but suffer from geometric inconsistencies during camera movements—objects morph, vanish, or distort unphysically because models lack 3D understanding. World-R1's key insight is to use reinforcement learning with a composite reward system that leverages pre-trained 3D models (Depth Anything 3) to reconstruct scenes as 3D Gaussian splats and penalize geometric inconsistencies through analysis-by-synthesis. The approach embeds camera trajectories directly into latent noise rather than modifying architectures, and uses Flow-GRPO to optimize the model based on 3D reconstruction quality, meta-view plausibility (evaluated by VLMs), and trajectory alignment.

Key Formula

Composite Reward Function

$$R(x, c) = R_{3D}(x, E, c) + \lambda_{gen} R_{gen}(x, c)$$

R(x,c) is total reward for generated video x given prompt c. R_{3D} is 3D-aware reward measuring geometric consistency, R_{gen} is general quality reward, E is camera trajectory, λ_{gen} is balancing hyperparameter. This combines 3D geometric fidelity with visual quality to guide RL optimization.

3D-Aware Reward Decomposition

$$R_{3D} = S_{meta} + S_{recon} + S_{traj}$$

R_{3D} decomposes into three components: S_{meta} evaluates meta-view plausibility using VLMs to detect geometric artifacts, S_{recon} measures reconstruction fidelity via LPIPS between original and re-rendered video, S_{traj} assesses camera trajectory alignment. This multi-faceted approach ensures comprehensive 3D consistency.

Key Figure

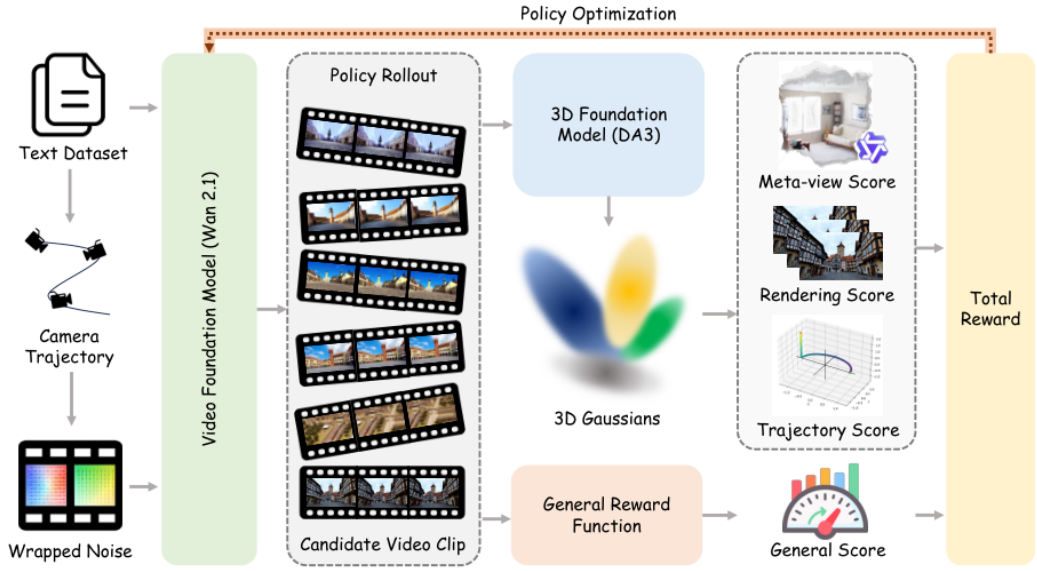

Figure 2 **Overview of World-R1.** *Left:* Given a text prompt, we synthesize a camera trajectory and embed it into the latent space via noise wrapping [34] to achieve implicit camera conditioning. *Middle:* The foundation model [3] generates candidate video clips. *Right:* We employ an comprehensive reward system: the generated video is lifted to a 3D Gaussian Splatting representation via a 3D Foundation Model [17]. We compute a 3D-aware reward score based on meta-view evaluation, rendering fidelity, and trajectory alignment, combined with a general generation reward. The model is optimized using Flow-GRPO-Fast [24] to internalize these geometric constraints.

Figure 2: Shows the complete World-R1 pipeline from text prompt to 3D-consistent video generation, illustrating the camera conditioning via noise wrapping, 3D reconstruction through Depth Anything 3, and multi-component reward system.

Limitations

- High computational cost of RL training requiring repeated video rollouts and reward evaluation compared to supervised fine-tuning
- Bounded by capabilities of underlying video foundation models, inheriting artifacts in challenging cases like dense multi-object scenes
- Periodic decoupled training adds complexity to balance geometric consistency with dynamic scene generation
- Reliance on pre-trained 3D foundation models and VLMs for reward computation introduces external dependencies